Multiview Visualisation of Uber Ride Booking Behaviour Data

Dataset link: *https://www.kaggle.com/datasets/yashdevladdha/uber-ride-analytics-dashboard*

Section 1 — The Data

The dataset used in this project contains ride-booking records from a ride-hailing platform. Each record represents a single booking transaction that includes information about ride characteristics, booking outcomes, vehicle types, locations, and payment methods. The dataset consists of 150,000 records and 21 attributes, providing a rich source of information for analysing patterns in urban mobility and ride-hailing demand.

```
(150000, 21)
Index(['Date', 'Time', 'Booking ID', 'Booking Status', 'Customer ID',
       'Vehicle Type', 'Pickup Location', 'Drop Location', 'Avg VTAT',
       'Avg CTAT', 'Cancelled Rides by Customer',
       'Reason for cancelling by Customer', 'Cancelled Rides by Driver',
       'Driver Cancellation Reason', 'Incomplete Rides',
       'Incomplete Rides Reason', 'Booking Value', 'Ride Distance',
       'Driver Ratings', 'Customer Rating', 'Payment Method'],
      dtype='object')
         Date      Time    Booking ID   Booking Status   Customer ID  \
0  23/03/2024  12:29:38  "CNR5884300"  No Driver Found  "CID1982111"
1  29/11/2024  18:01:39  "CNR1326809"       Incomplete  "CID4604802"
2  23/08/2024  08:56:10  "CNR8494506"        Completed  "CID9202816"
3  21/10/2024  17:17:25  "CNR8906825"        Completed  "CID2610914"
4  16/09/2024  22:08:00  "CNR1950162"        Completed  "CID9933542"

     Vehicle Type      Pickup Location      Drop Location  Avg VTAT  Avg CTAT  \
0          eBike          Palam Vihar            Jhilmil       NaN       NaN
1        Go Sedan         Shastri Nagar  Gurgaon Sector 56       4.9      14.0
2           Auto             Khandsa       Malviya Nagar      13.4      25.8
3  Premier Sedan  Central Secretariat           Inderlok      13.1      28.5
4           Bike     Ghitorni Village        Khan Market       5.3      19.6

   ...  Reason for cancelling by Customer Cancelled Rides by Driver  \
0  ...                                NaN                       NaN
...
 20  Payment Method                     102000 non-null  object
dtypes: float64(9), object(12)
memory usage: 24.0+ MB
None
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Dataset Information

Each row in the dataset corresponds to an individual ride booking event. The dataset includes both completed rides and unsuccessful booking attempts, allowing analysis of ride demand as well as booking outcomes such as cancellations or incomplete rides. The attributes in the dataset can be categorised using common information visualisation data types.

```
Missing values per column:
Date                                       0
Time                                       0
Booking ID                                 0
Booking Status                             0
Customer ID                                0
Vehicle Type                               0
Pickup Location                            0
Drop Location                              0
Avg VTAT                               10500
Avg CTAT                               48000
Cancelled Rides by Customer           139500
Reason for cancelling by Customer     139500
Cancelled Rides by Driver             123000
Driver Cancellation Reason            123000
Incomplete Rides                      141000
Incomplete Rides Reason               141000
Booking Value                          48000
Ride Distance                          48000
Driver Ratings                         57000
Customer Rating                        57000
Payment Method                         48000
dtype: int64
...
Uber Wallet    12276
Credit Card    10209
Debit Card      8239
Name: count, dtype: int64
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Missing Column Counts

Temporal attributes describe when the booking occurred. These include the variables *Date* and *Time*. Additional temporal features such as *Hour*, *Weekday*, and *Month* were derived during preprocessing in order to enable temporal analysis of ride demand patterns.

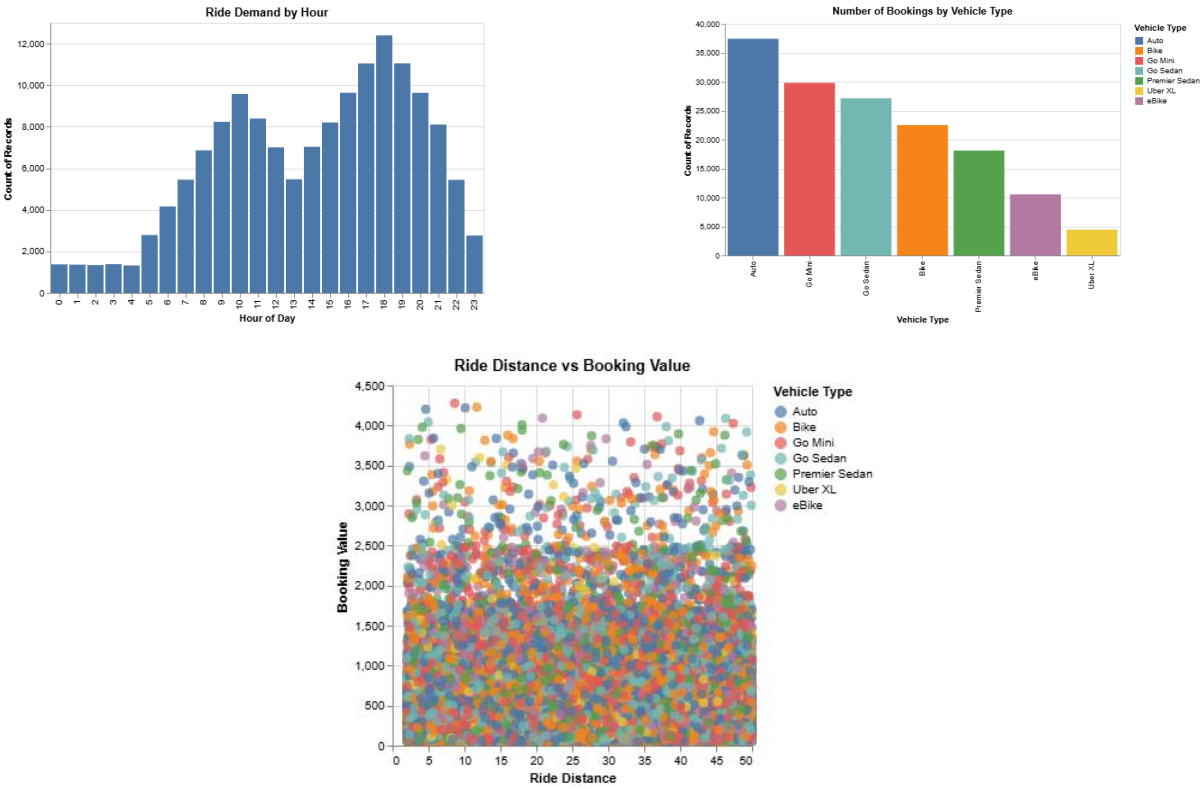
Categorical attributes represent discrete groups or categories. Important categorical variables include *Vehicle Type*, *Booking Status*, *Pickup Location*, *Drop Location*, and *Payment Method*. These variables allow comparisons across categories such as vehicle classes or booking outcomes.

Quantitative attributes represent measurable numerical values associated with rides. These include *Ride Distance*, *Booking Value*, *Driver Ratings*, and *Customer Rating*. These attributes enable analysis of relationships between ride characteristics such as distance and cost.

Identifier attributes such as *Booking ID* and *Customer ID* uniquely identify individual transactions or users. While these attributes are important for data integrity, they are not directly used in visualisation tasks.

Several attributes contain missing values because they only apply to specific ride outcomes. For example, *Ride Distance* and *Booking Value* are only available for completed rides, while cancellation reasons only appear for cancelled bookings. Therefore, some analyses in this project use filtered subsets of the dataset, such as completed rides only.

Overall, the dataset contains a mixture of temporal, categorical, and quantitative attributes, making it suitable for interactive multi-view visualisation and exploratory analysis of ride-hailing behaviour.



Some Visualizations from the Dataset

Section 2 — The Tasks

In order to support exploratory analysis of the ride-booking dataset, a set of user tasks was defined based on common information visualisation task taxonomies. These tasks represent typical analytical actions that users may perform when exploring ride-hailing data. Each of the visualisation systems developed in this project supports all defined tasks.

<u>T1 – Compare ride demand across vehicle types.</u>
Users examine how booking demand differs between vehicle categories such as Auto, Go

Mini, Go Sedan, Bike, and other vehicle types. This task involves comparing aggregated booking counts across categorical groups in order to identify the most frequently used transportation options.

T2 – Analyse temporal patterns in ride demand.
Users explore how ride demand changes throughout the day by analysing the distribution of bookings across hourly time intervals. This task enables identification of peak demand periods and patterns related to commuting behaviour.

T3 – Explore relationships between ride distance and booking value.
Users investigate the relationship between ride distance and booking value in order to understand pricing behaviour and ride characteristics. This task involves analysing quantitative relationships between two continuous variables.

T4 – Investigate booking outcomes and ride completion patterns.
Users analyse how booking outcomes vary across the dataset by examining different booking statuses such as completed rides, cancellations by drivers, cancellations by customers, and incomplete rides. This task enables exploration of ride completion reliability and booking success rates.

T5 – Select and explore subsets of rides.
Users interactively filter the dataset in order to analyse specific subsets of rides based on attributes such as vehicle type, pickup location, drop location, or time period. This task supports detailed exploration of particular segments of the dataset and enables focused analysis of user-defined selections.

Together, these tasks support comparison, trend analysis, relationship exploration, and interactive selection of subsets within the dataset. The defined tasks provide the analytical foundation for the three visualisation systems developed in this project.

Section 3 — The Core Systems

System A: Demand Overview Dashboard

System A is designed as a foundational, standard dashboard that provides an intuitive overview of ride-hailing demand patterns. It uses classic, highly readable chart types to ensure users can quickly grasp absolute values and general distributions across the dataset. The system is a multi-view composition consisting of four visualizations, successfully integrating all predefined user tasks (T1-T5) through interactive unidirectional linking.

Vehicle Type Demand (T1): A vertical bar chart displays the total number of bookings categorized by vehicle type. This encoding uses length to accurately represent and compare categorical aggregates, providing a clear ranking of transportation preferences.

Temporal Patterns (T2): A line chart plots the total bookings against the hour of the day. The continuous nature of the line encoding effectively highlights cyclical temporal trends and peak commuting hours.

Distance vs. Value Relationship (T3): A scatter plot maps ride distance on the x-axis against booking value on the y-axis. To address the visual compression caused by extreme outliers, the axes' domains are clamped, and interactive geometric zooming and panning are enabled. Combined with reduced point opacity, this view allows users to flexibly explore normal pricing patterns while still being able to identify extreme data points.

Booking Outcomes (T4): A sorted vertical bar chart categorizes the absolute counts of different booking statuses (e.g., Completed, Cancelled, Incomplete). This allows for a straightforward comparison of successful rides versus failures.

Interactive Subset Selection (T5): System A employs unidirectional brushing and linking. The Vehicle Type bar chart (T1) serves as the interactive control interface. When a user clicks on a specific vehicle category (e.g., "Go Sedan"), a selection parameter dynamically filters the underlying dataset. Consequently, the line chart (T2), scatter plot (T3), and status bar chart (T4) instantly update to reflect only the data associated with the selected vehicle type. This interaction empowers users to drill down into the behavioral differences of specific transportation subsets.

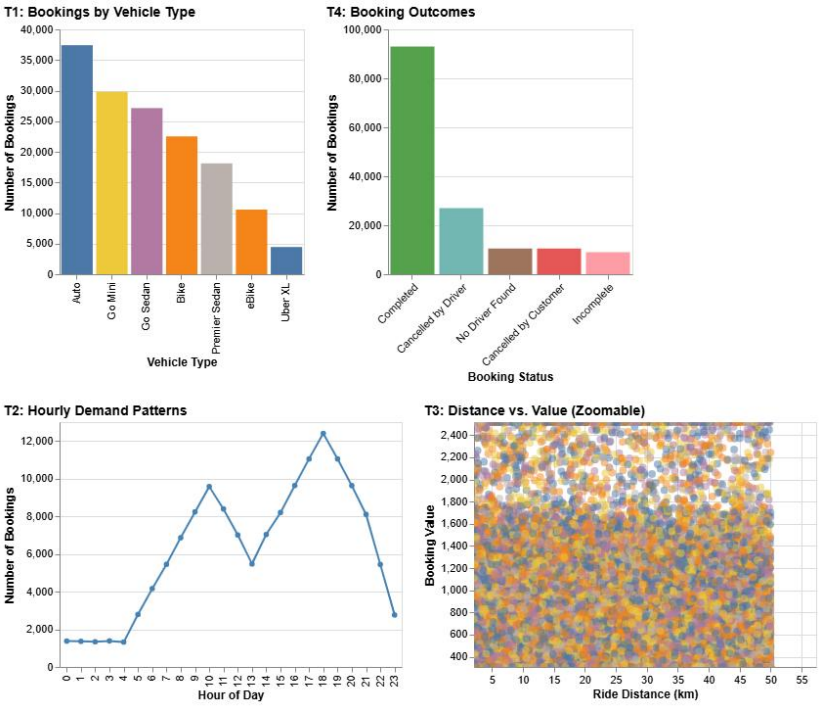


Figure 1: System A

System B: Density and Time Characterisation System

System B is designed to provide a deeper analytical view of ride-hailing booking behaviour by transforming individual ride records into aggregated density and hierarchical temporal visualisations. Instead of showing every data point directly, the system highlights patterns in the data distribution and allows users to explore ride demand at multiple temporal levels. The system contains three interconnected visualisations and uses interactive filtering techniques to support all predefined tasks (T1–T5).

The first view visualises the relationship between ride distance and booking value using a two-dimensional heatmap. Unlike the scatterplot used in System A, the heatmap aggregates individual data points into a 40×40 grid. Each grid cell represents the number of rides occurring within that distance–value range, with colour intensity indicating the density of records. This approach reduces visual clutter caused by overlapping points and allows users to easily identify the most common pricing patterns and ride characteristics.

The second view is a stacked bar chart showing the number of rides for each hour of the day.

Different booking outcomes are represented using colours from the Tableau10 palette, allowing users to observe both the temporal demand pattern and the distribution of booking statuses simultaneously. Through this view, users can analyse peak booking periods (T2) and compare the proportions of successful and unsuccessful bookings across different hours (T4).

The third view acts as an interaction panel that groups rides into four broader time categories: Night (0–5), Morning (6–11), Afternoon (12–17), and Evening (18–23). This bar chart provides a high-level summary of demand across major time periods and serves as a control interface for generalised selection.

System B supports coordinated interaction between views. Users can brush a rectangular region in the density heatmap to select rides with specific distance and booking value ranges. This selection dynamically filters the hourly bar chart to show only rides within the selected region. In addition, selecting a time category in the control panel highlights the corresponding hours and updates the other views accordingly. This design allows users to smoothly move between detailed ride data and broader temporal patterns.

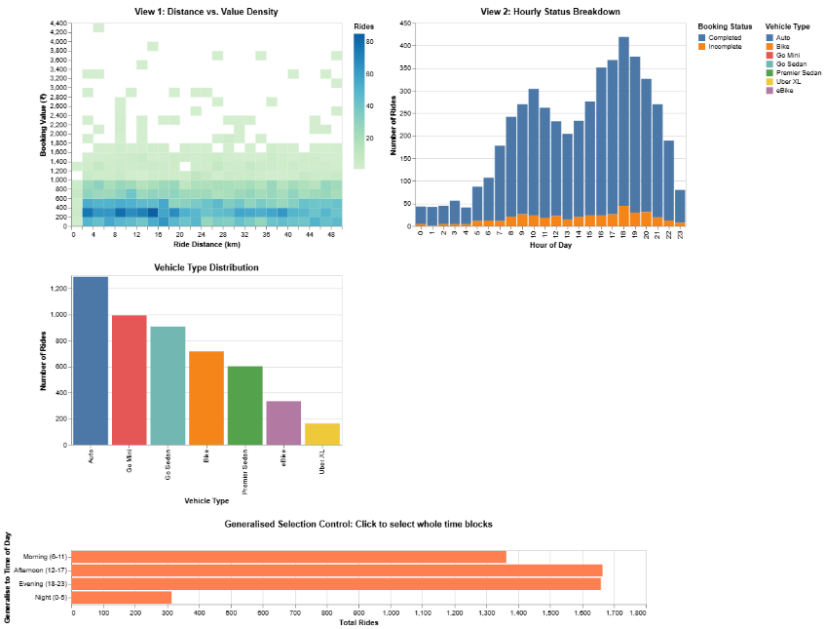


Figure 2: System B

System C – Spatial and Temporal Ride Demand Dashboard

System C is designed as a spatial exploration dashboard that focuses on understanding where rides originate, where they end, and how ride demand changes over time. The system combines three linked visualisations to demonstrate spatial travel patterns and temporal behaviour. It supports all predefined analytical tasks (T1–T5) through interactive filtering with brushing and linking.

The first view presents a horizontal stacked bar chart showing the twenty most frequent pickup locations. Each bar represents the total number of rides originating from a location, while colour encodes booking status categories such as Completed, Cancelled by Driver, Cancelled by Customer, and Incomplete. This design allows users to analyse both spatial demand and booking reliability simultaneously. By comparing the coloured segments within each bar, users can observe how booking outcomes vary across different origin locations, supporting task T4. The second view displays the twenty most common drop locations using a horizontal bar chart. This chart dynamically updates when users select one or more pickup

locations in the first view. The interaction enables exploration of origin–destination relationships by showing where rides from a selected location most frequently end. Both of these views support multiselection.Temporal demand patterns are represented using a heatmap that plots the hour of the day on the horizontal axis and the day of the week on the vertical axis. Colour intensity indicates the number of rides occurring during each time interval, enabling users to identify peak travel periods and weekly demand cycles. When a pickup location, drop location, or vehicle type is selected, the heatmap updates to reflect the temporal demand patterns associated with the selected subset of rides, supporting task T2.

System C also includes a vehicle type dropdown filter that allows users to restrict the dataset to a specific transportation category. This enables comparison of spatial demand patterns across vehicle types, supporting task T1. Together with the interactive pickup selection, these filters allow users to explore subsets of rides across multiple views, supporting task T5. Although System C does not directly visualise the relationship between ride distance and booking value (T3), contextual information about these variables is provided through heatmap tooltips displaying average booking value and average ride distance for each time interval.

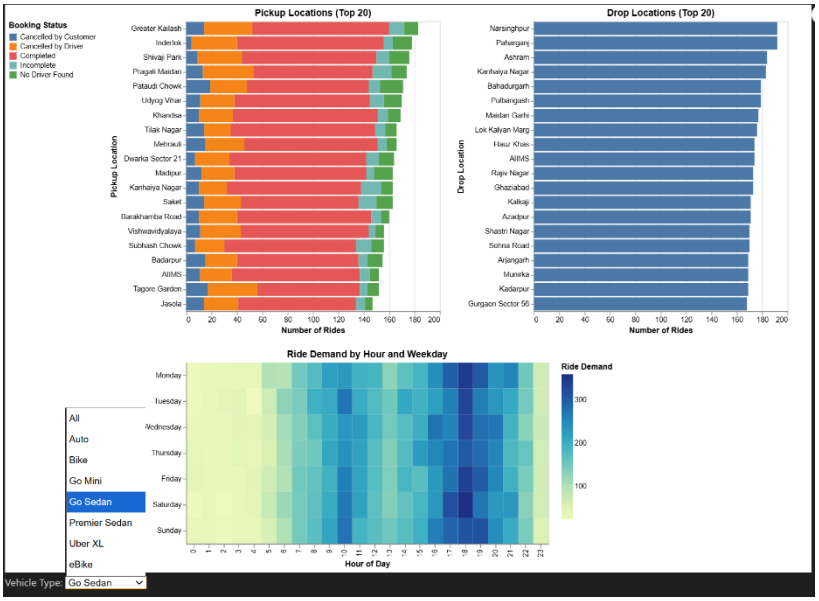


Figure 3: System C

Section 4 — Generalised Selection

Generalised selection is implemented in **System B** to allow users to explore ride bookings at different levels of temporal abstraction. Instead of selecting only individual data points, the system enables users to select broader groups of rides through a hierarchical structure based on time.

Semantic Structure

The semantic hierarchy used in this system is based on the **time of day**. At the lowest level of the hierarchy, each ride is associated with a specific **hour of the day (0–23)** derived from the booking time. These hours are then grouped into four broader **time-of-day categories**:

- Night (0–5)
- Morning (6–11)
- Afternoon (12–17)
- Evening (18–23)

This hierarchy allows the data to be interpreted at two different levels: a **detailed hourly level** and a **more general time-of-day level**.

Traversal Policy

The traversal policy defines how the user moves from a specific selection to a more general group of data. In this system, the user can generalise a selection by choosing one of the broader time-of-day categories. When a user selects a time block such as **Morning**, the system automatically selects all rides that occurred between **6:00 and 11:59**. This allows the selection to move from individual hours to a larger semantic category.

Through this interaction, users can quickly analyse patterns across larger temporal groups instead of inspecting each hour individually.

Implementation

The implementation of generalised selection is achieved by creating a derived attribute called **Time_of_Day**, which assigns each hour value to one of the four temporal categories. This attribute is used in the **generalised selection control panel**, shown as a horizontal bar chart at the bottom of System B.

Each bar represents the total number of rides within a time-of-day category. When a user clicks on one of these bars, the selection filters the other views in the system. The density heatmap and hourly stacked bar chart update to display only rides belonging to the selected time period.

This approach allows users to smoothly move between detailed ride behaviour and broader daily patterns, demonstrating how generalised selection can support multi-level data exploration.

Section 5 — Demo Video

A video demonstration of all three visualisation systems (A, B, and C), including their design, implementation, and interaction features, is available at the link provided at the beginning of this document.

Section 6 — Design Comparison

This section compares the three visualisation systems by analysing six design decisions made during the development process. Each decision reflects a different approach taken by Systems A, B, and C to support the tasks defined earlier. The comparison highlights how different visualisation techniques influence usability and analytical effectiveness.

Design Decision 1 – Visualising ride demand by vehicle type (T1)

System A uses a bar chart to display the number of bookings for each vehicle type. This approach allows users to easily compare categories because bar length is a clear and accurate

encoding for categorical comparisons. **System B** shows vehicle type distribution using a filtered bar chart combined with density exploration. The system focuses more on ride behaviour patterns rather than simple categorical comparisons. **System C** supports this task using a dropdown vehicle type filter, allowing users to examine how spatial patterns change for different vehicle types.

Among the three approaches, **System A** provides the clearest solution for this task because the bar chart allows direct visual comparison of vehicle demand across categories. The designs used in Systems B and C support the task indirectly through filtering rather than direct comparison.

Design Decision 2 – Representing temporal ride demand (T2)

System A uses a line chart to visualise booking counts across the hours of the day. This design emphasises trends and makes it easy to identify peak demand periods**. System B** uses a stacked bar chart by hour, where each bar is divided according to booking status. This allows users to analyse both demand and ride outcomes at the same time**. System C** uses a heatmap showing hour versus weekday, where colour intensity represents ride demand. This allows users to analyse daily and weekly temporal patterns at the same time.

Among these approaches, **System C** provides the richest temporal insight, because it reveals both hourly and weekly patterns simultaneously. However, System A is the simplest to interpret for quick trend identification.

Design Decision 3 – Representing the relationship between ride distance and booking value (T3)

System A uses a scatter plot to show individual rides with ride distance on the x-axis and booking value on the y-axis. This design makes it possible to observe the distribution of individual data points. **System B** replaces the scatter plot with a two-dimensional density heatmap. The heatmap aggregates points into bins, which reduces visual clutter when large numbers of rides overlap. **System C** does not display this relationship directly but provides contextual information through tooltip summaries of average distance and booking value in the heatmap.

Among these approaches, **System B** is the most effective, because the density heatmap handles large datasets better and makes common pricing patterns easier to identify.

Design Decision 4 – Interaction techniques and filtering (T5)

System A uses simple click-based filtering. Selecting a vehicle type updates the other charts to show only rides belonging to that category. **System B** implements brushing and generalised selection, allowing users to draw a selection region in the heatmap and also choose broader time-of-day categories. **System C** uses multiple linked filters, including pickup selection, drop selection, and a vehicle type dropdown.

Among these approaches, **System B** provides the most advanced interaction design, because it combines brushing with semantic hierarchical selection. This allows both detailed and generalised exploration of the data.

Design Decision 5 – Handling high-density datasets

System A visualises individual data points with reduced opacity in the scatter plot to reduce overlap. **System B** uses binning and aggregation, transforming large numbers of points into a density heatmap. **System C** uses aggregation through bar charts and heatmaps, which summarise the data rather than showing individual records.

Among these approaches, **System B** handles dense datasets most effectively, because binning reduces overplotting while still preserving meaningful data patterns.

Design Decision 6 – Representing spatial ride behaviour

System A does not include spatial information and focuses mainly on demand and ride characteristics. **System B** focuses on behavioural and temporal patterns but also does not directly visualise spatial relationships**. System C** introduces pickup and drop location charts, allowing users to explore origin–destination relationships between different areas of the city.

Among the three systems, **System C** provides the strongest support for spatial analysis, because it directly visualises where rides start and end and links this information with temporal demand patterns.

Design Comparison Summary

The comparison shows that each system has different strengths. System A provides a clear view of basic ride demand patterns and is easy to interpret. System B introduces more advanced analytical techniques such as density visualisation and generalised selection, making it effective for deeper analysis. System C focuses on spatial exploration and provides useful insights into origin–destination patterns and temporal demand. Together, the three systems demonstrate how different visualisation design choices can support the same analytical tasks in different ways.

Section 7 — User Evaluation

To evaluate the effectiveness of the three visualisation systems, a user study was conducted with multiple participants. The goal of this evaluation was to compare how well each system supports the predefined analytical tasks (T1–T5), and to measure usability in terms of task completion time, error rate, and user preference.

Evaluation Setup

A total of 5 participants were asked to complete the same set of five tasks using each of the three systems (System A, System B, and System C). The evaluation metrics were chosen to measure efficiency (time), accuracy (errors), and usability (preference), which are common criteria in visualisation evaluation. The tasks were designed based on the defined task set:

T1: Identify the most used vehicle type;

T2: Identify peak demand hours;

T3: Understand the relationship between ride distance and booking value;

T4: Identify the most common booking outcome;

T5: Interact with the system to filter and explore subsets of data;

Participants performed each task on all three systems. During the evaluation, three types of data were recorded:

-Task completion time (in seconds);

-Number of errors made during task execution;

-User preference rating on a scale from 1 (very difficult) to 5 (very easy);

The aggregated results of the evaluation are summarised below:

	Average task completion time	Total errors	Average user preference
System A	8.31 seconds	5	3.8/5
System B	12.23 seconds	9	3.20/5
System C	17.59 seconds	11	2.36/5

Results

Analysis

The results show clear differences in usability between the three systems.

System A achieved the best overall performance across all evaluation metrics. It had the fastest task completion time and the lowest number of errors, indicating that users were able to quickly understand and interact with the visualisations. The relatively high preference score also suggests that users found System A easy to use and intuitive. This can be explained by its use of simple and familiar chart types such as bar charts and line charts, as well as straightforward interaction through click-based filtering.

System B showed moderate performance. While it provides more advanced analytical capabilities, such as density visualisation and hierarchical time selection, it required more time for users to complete tasks and resulted in a higher number of errors compared to System A. This suggests that although System B is more powerful for deeper analysis, it introduces additional complexity that may reduce usability for inexperienced users. The slightly lower preference score reflects this increased cognitive load.

System C performed the weakest among the three systems. It had the highest task completion time, the highest number of errors, and the lowest user preference score. This indicates that users found it more difficult to navigate and interpret. The inclusion of spatial and multi-filter interactions, while useful for detailed exploration, may have made the system more complex

and less intuitive. In particular, users may have struggled with understanding the relationships between multiple linked views.

User Evaluation Conclusion

Overall, the evaluation demonstrates that there is a trade-off between simplicity and analytical power. System A is the most effective for quick and intuitive data exploration, while System B offers more advanced analytical capabilities at the cost of increased complexity. System C provides valuable spatial insights but suffers from lower usability due to its more complex interaction design.

These findings highlight the importance of balancing visual complexity and usability when designing interactive visualisation systems.

Section 8 — Future Work

This project demonstrates how different visualisation designs can support the exploration of ride-hailing data. However, several improvements could be made in future work.

First, the systems could be extended by incorporating real geographic maps instead of categorical location charts. This would allow users to better understand spatial patterns and navigation behaviour across the city. Second, additional interaction techniques could be introduced to improve usability. For example, tooltips could be enhanced with more detailed summaries, and guided interactions could help users better understand complex visualisations such as the density heatmap in System B. Third, more advanced data analysis methods could be integrated. For instance, clustering or machine learning techniques could be used to identify hidden patterns in ride behaviour, such as grouping similar trip types or detecting anomalies. Finally, performance optimisation could be considered for handling larger datasets in real time. This would make the systems more scalable and suitable for real-world applications.

Overall, these improvements would enhance both the analytical power and usability of the visualisation systems.

Appendix A — User Evaluation Data

TaskNumber	Participant	System	AvgTime	Errors	Preference
1	P1	A	3.31	0	4
1	P1	B	1.98	0	5
1	P1	C	20.53	1	0
1	P2	A	2.31	0	5
1	P2	B	2.68	0	5
1	P2	C	23.89	1	0
1	P3	A	2.66	0	5
1	P3	B	5.5	0	3
1	P3	C	34.5	2	1
1	P4	A	2.97	0	4
1	P4	B	3.12	0	5
1	P4	C	22.18	1	0

1	P5	A	1.9	0	4
1	P5	B	7.4	0	5
1	P5	C	20.78	1	1
2	P1	A	10.16	0	3
2	P1	B	4.75	0	5
2	P1	C	9.25	0	4
2	P2	A	4.28	0	4
2	P2	B	2.92	0	5
2	P2	C	5.21	0	4
2	P3	A	7.07	0	5
2	P3	B	9.19	0	4
2	P3	C	8.53	0	4
2	P4	A	6.33	0	4
2	P4	B	5.17	0	5
2	P4	C	7.89	0	4
2	P5	A	6.43	0	4
2	P5	B	6.78	0	4
2	P5	C	8.47	0	3
3	P1	A	25.83	1	0
3	P1	B	22.61	1	2
3	P1	C	15.76	1	0
3	P2	A	28.1	1	0
3	P2	B	19.22	1	2
3	P2	C	15.93	1	0
3	P3	A	14.34	1	1
3	P3	B	25.72	0	4
3	P3	C	30.23	1	2
3	P4	A	22.47	1	1
3	P4	B	20.35	1	3
3	P4	C	18.62	1	1
3	P5	A	21.54	1	2
3	P5	B	20.46	1	3
3	P5	C	22.73	1	1
4	P1	A	2.5	0	5
4	P1	B	10.1	0	4
4	P1	C	11.31	0	2
4	P2	A	2.13	0	5
4	P2	B	5.74	0	4
4	P2	C	16.41	0	3
4	P3	A	2.86	0	5
4	P3	B	30.24	0	3
4	P3	C	15.29	0	4
4	P4	A	2.69	0	5
4	P4	B	7.22	0	4
4	P4	C	13.87	0	3
4	P5	A	2.78	0	5
4	P5	B	8.37	0	4
4	P5	C	17.83	0	2
5	P1	A	4.78	0	5
5	P1	B	4.2	1	0

5	P1	C	27.52	0	4
5	P2	A	3.83	0	4
5	P2	B	21.01	1	0
5	P2	C	17.31	0	3
5	P3	A	14.52	0	5
5	P3	B	27.43	1	0
5	P3	C	14.51	0	5
5	P4	A	5.11	0	5
5	P4	B	19.88	1	0
5	P4	C	21.45	0	4
5	P5	A	6.93	0	5
5	P5	B	13.7	1	1
5	P5	C	19.8	0	4

Reference List

Visualization Grammars and Altair/Vega-Lite

[1] **Satyanarayan, A., Moritz, D., Wongsuphasawat, K. and Heer, J.** (2017). *Vega-Lite: A Grammar of Interactive Graphics.* IEEE Transactions on Visualization and Computer Graphics, 23(1), pp. 341–350. DOI: 10.1109/TVCG.2016.2599030 Available at: https://dl.acm.org/doi/10.1109/TVCG.2016.2599030

> **Relevance:** The foundational paper for Vega-Lite, covering selections, brushing, linking, and the composition algebra used across all three systems.

[2] **VanderPlas, J., Granger, B., Heer, J., Moritz, D., Wongsuphasawat, K. and Satyanarayan, A.** (2018). *Altair: Interactive Statistical Visualizations for Python.* Journal of Open Source Software, 3(32), 1057. DOI: 10.21105/joss.01057 Available at: https://dig.cmu.edu/publications/2018-altair.html

> **Relevance:** The paper on the Altair Python library used to build all three systems.

[3] **Wilkinson, L.** (2005). *The Grammar of Graphics.* 2nd ed. New York: Springer.

> **Relevance:** The seminal book that established the grammar of graphics framework underlying Vega-Lite and Altair.

Interactive Visualization Design Principles

[4] **Shneiderman, B.** (1996). *The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations.* Proceedings of the IEEE Symposium on Visual Languages, pp. 336–343. Available at: https://ieeexplore.ieee.org/document/545307

> **Relevance:** Defines the "Overview first, zoom and filter, then details-on-demand" mantra, directly relevant to the dashboard designs, especially System B's generalised selection control.

[5] **Munzner, T.** (2014). *Visualization Analysis and Design.* A K Peters/CRC Press. Available at: https://www.cs.ubc.ca/~tmm/vadbook/

> **Relevance:** A comprehensive framework for marks, channels, multiple views, and focus+context, applicable to justifying encoding choices (bar charts, heatmaps, scatter plots).

Coordinated Multiple Views and Brushing & Linking

[6] **Roberts, J.C.** (2007). *State of the Art: Coordinated & Multiple Views in Exploratory Visualization.* Proceedings of the Fifth International Conference on Coordinated and Multiple Views in Exploratory Visualization (CMV '07), pp. 61–71. DOI: 10.1109/CMV.2007.20 Available at: https://www.cs.kent.ac.uk/pubs/2007/2559/content.pdf

> **Relevance:** A key survey on CMV techniques, directly supporting the brushing and linking interactions in all three systems.

[7] **Wang Baldonado, M.Q., Woodruff, A. and Kuchinsky, A.** (2000). *Guidelines for Using Multiple Views in Information Visualization.* Proceedings of the Working Conference on Advanced Visual Interfaces (AVI '00), pp. 110–119. DOI: 10.1145/345513.345271 Available at: https://dl.acm.org/doi/10.1145/345513.345271

> **Relevance:** Provides design guidelines for when and how to use multiple coordinated views effectively.

Time-Oriented and Temporal Visualization

[8] **Aigner, W., Miksch, S., Schumann, H. and Tominski, C.** (2011). *Visualization of Time-Oriented Data.* London: Springer. Available at: https://timeviz.net

Relevance: Relevant to the hourly demand patterns (System A's line chart, System B's stacked bar chart, System C's heatmap by hour and weekday).

Ride-Hailing and Transportation Data Analysis

[9] **IJRASET** (2025). *Exploratory Data Analytics on Uber Transportation Patterns.* International Journal for Research in Applied Science and Engineering Technology. Available at: https://www.ijraset.com/research-paper/exploratory-data-analytics-on-uber-transportation-patterns

Relevance: A comparable study using Uber trip data for demand pattern visualization and analysis.

[10] **Nano-NTP** (2024). *Comprehensive Analysis of Uber Ride Patterns.* Nanotechnology Perceptions, 20(S14), pp. 4740–4757. Available at: https://nano-ntp.com/index.php/nano/article/download/5659/4547/11113

Relevance: Covers EDA on Uber ride data including temporal analysis, spatial analysis, and demand peak identification.

[11] **IEEE** (2025). *Transportation Card Data Analysis for Ride-Hailing Demand Estimation.* IEEE Xplore. Available at: https://ieeexplore.ieee.org/document/10825766/

Relevance: Explores ride-hailing demand estimation approaches relevant to booking and demand analysis.

Semantic Zooming and Generalised Selection

[12] **Wiens, V., Lohmann, S. and Auer, S.** (2017). *Semantic Zooming for Ontology Graph Visualizations.* Proceedings of the Knowledge Capture Conference (K-CAP '17). DOI: 10.1145/3148011.3148015 Available at: https://dl.acm.org/doi/10.1145/3148011.3148015

Relevance: Relevant to System B's generalised selection concept, where semantic hierarchy (Hour → Time of Day) is implemented.